

A3: Proposed Enhancement

finneganmac.github.io/CISC322-A2-Presentation.mp4

GEMINI

Tips for getting started:
1. Ask questions, edit files, or run commands.
2. Be specific for the best results.
3. Help for more information.

Write a short paragraph about why Gemini CLI is awesome

The Rat Pack

Rémy Brown-Vandervecht

Jude Henderson – Group Leader

Caleb Gao

Finnegan MacAdams – Presenter

Cory Boys - Presenter

Aatman Gupta



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Audio Feature

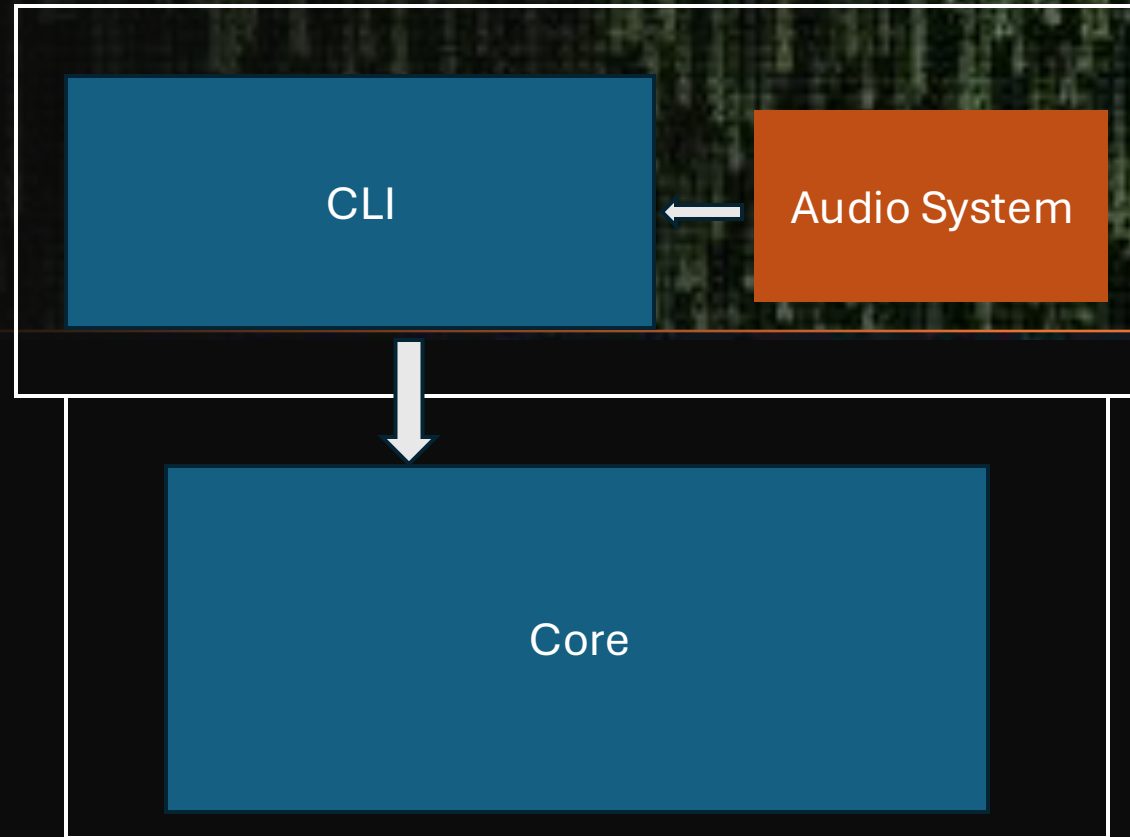


- Speech to text
- Text to Speech

For user accessibility

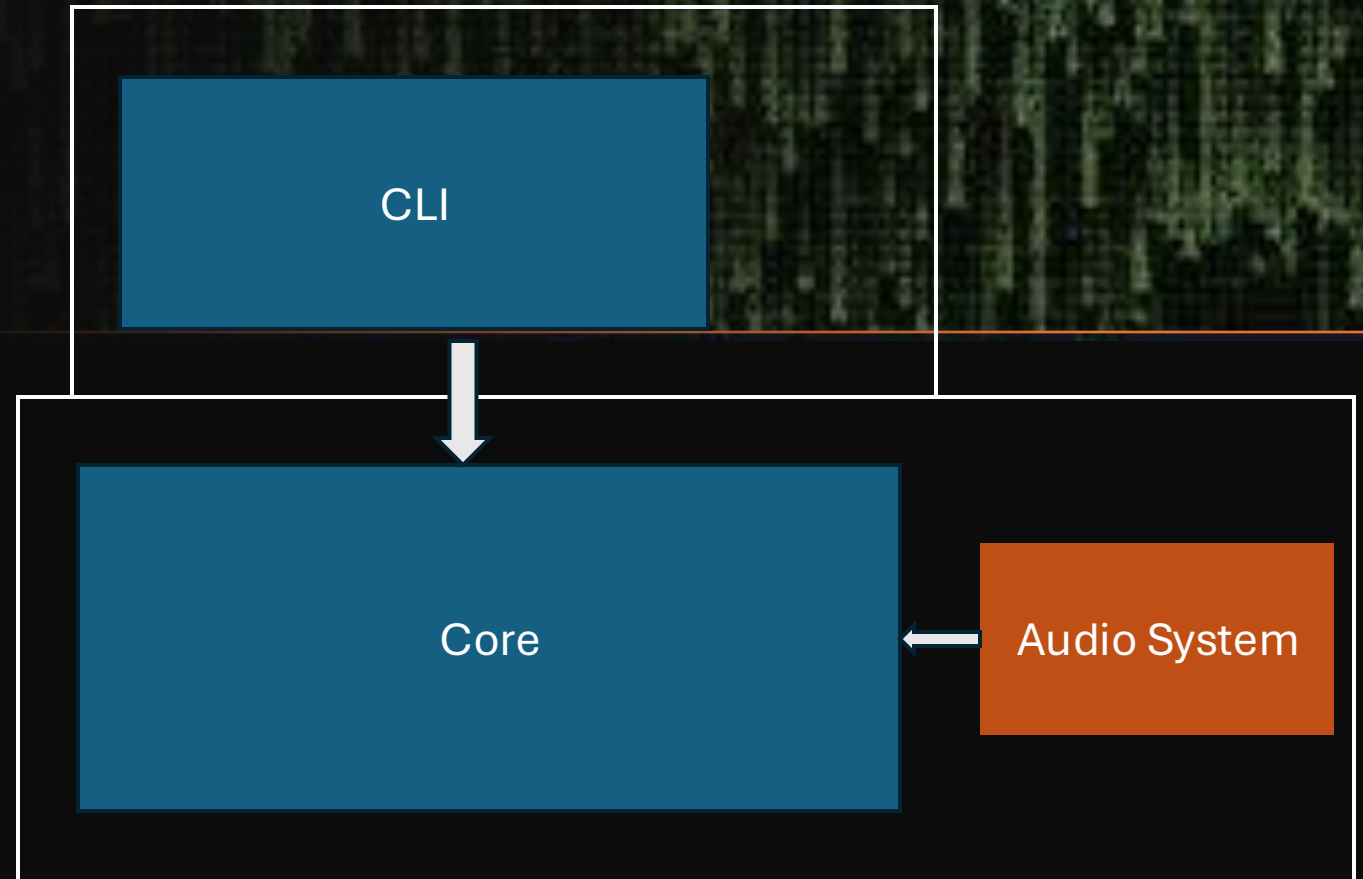
Architecture Implementation #1: Local Processing

- Audio Processing connected to the CLI package
- Processed within the user's device
- Separated from system backend



Architecture Implementation #2: Remote server processing

- Audio Processed remotely on a cloud API
- Interacting with the Core package



Evaluation of Alternatives

Privacy

#1 has better security

- Not travelling across a network
- No third-party data retention

Performance

#2 has better performance

- Offloads computation to the cloud
- Could have network latency

Maintainability

#2 Doesn't have OS specific dependencies

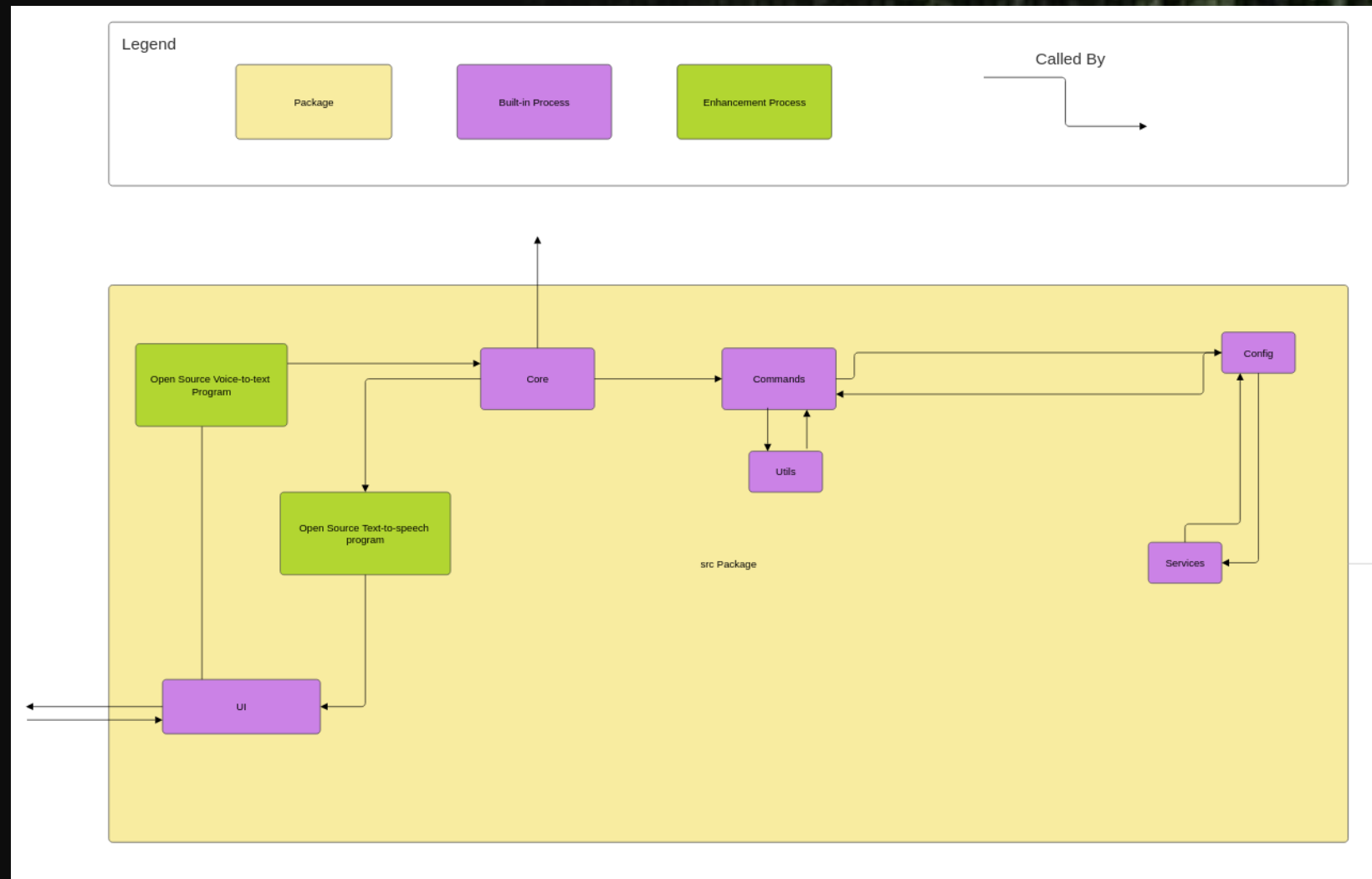
- No code changes required

Usability

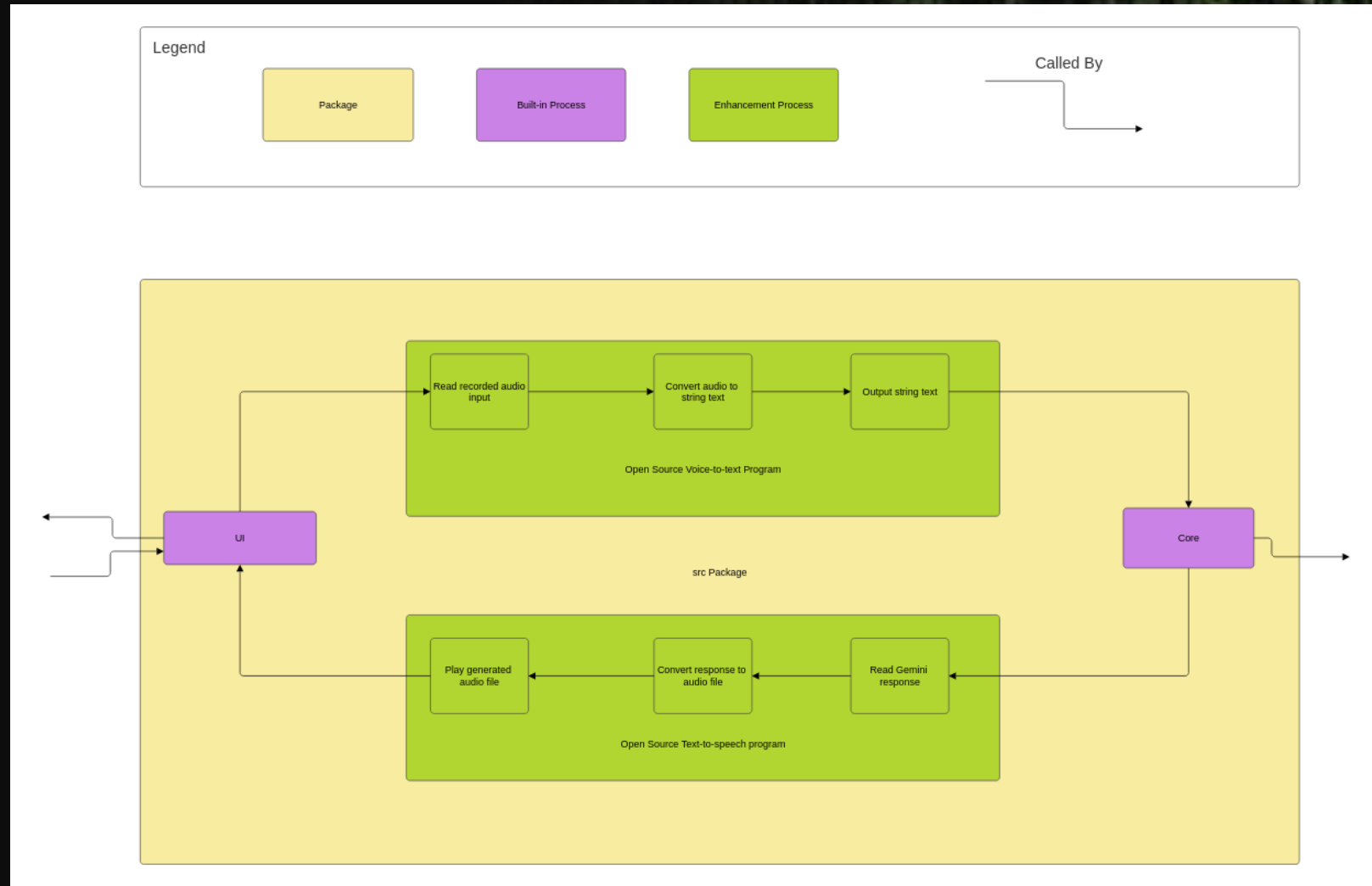
#2 has better usability

- Cloud providers trained on bigger datasets
- Better for programming syntax

Impacted Subsystem #1



Impacted Subsystem #2



Plan for Testing Enhancement



Timer & Resource monitor

- Measure RAM and CPU usage
- User wait time for playable audio file

Potential Risks

Security

- Background noise could be picked up
- Audio in external cloud service

Performance

- Already has intensive background processes
- Microphone polling and audio buffering can add to that

Maintainability

- Audio processing variability across OSes
- Needs testing on all 3 platforms

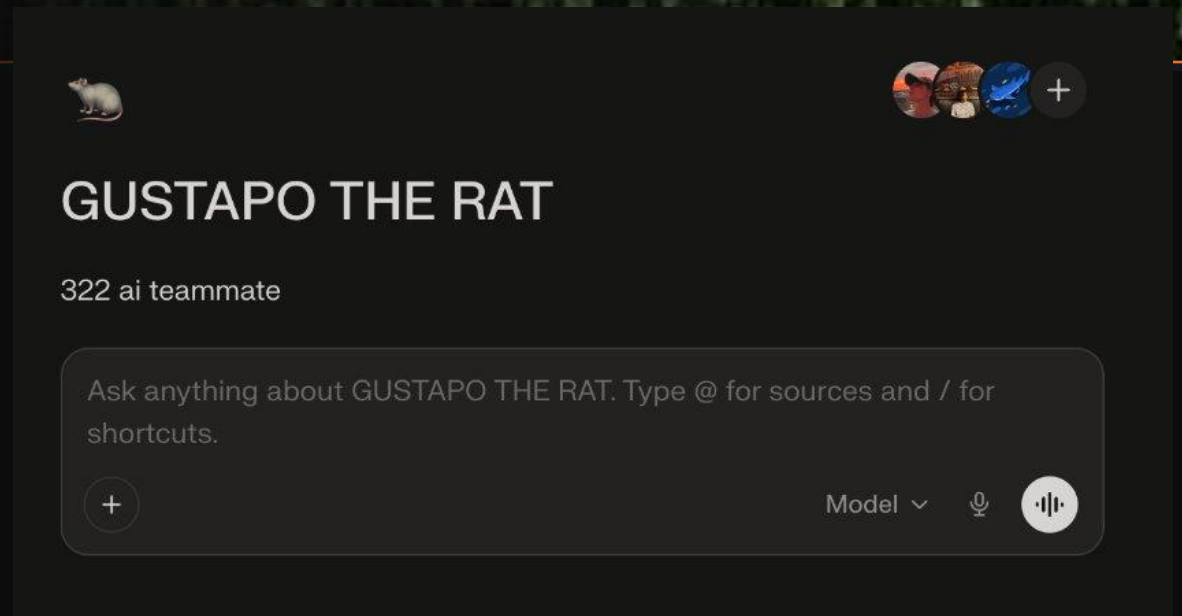
Usability

- Transcription errors
- Highly specific syntax
- Train API on programming specific commands

AI Usage - Gustapo

Perplexity AI – Gustapo the Rat

- All interactions restricted to a shared space
- Uploaded important, course-specific documents
- Gave feedback based on rubric



Conclusion

- Compared local host VS remote server for audio processing
- Considered possible risks associated with chosen implementation
- Potential solutions to these risks

Overall improvement in accessibility of the system

